

COSMOS-P03

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

**Method of family-axis occupancy using quantified porosity and
different-mistake overlay****ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P03**Kind:** Method / occupancy rule**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree
discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click
USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35
U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of
a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

*This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and
not a novelty opinion.*

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to seating generative models for a task by family and by quantified mistake surface (porosity), so that different-mistake models overlay residual holes faster than copies of the same family.

BACKGROUND OF THE INVENTION

[0003] N copies of one API bill N times and keep the same holes. Self-consistency and more-agents-is-all-you-need stack copies. Hidden Clones (2026) and LLMs-as-Jury (2026) already measure family-correlated errors. Knight and Leveson 1986: independently written versions do not fail independently.

[0004] The inventor names the quality of a coder as porosity (spoken Pouroosity): the hole-set or mistake surface of that model (error classes, unseen tools, fabrication modes). Low-quality models with different mistakes, when overlaid, cover surface quickly at a greater-than-linear rate versus stacking copies. Copies of the same family leave holes aligned. This disclosure does not invent an exponent. Convert with a measurement.

BRIEF SUMMARY OF THE INVENTION

[0005] Assign each model a family (training house, tool surface, observed failure mode). Quantify each coder's porosity as a hole-set measure. When seating votes, the same family does not count twice. Overlay: residual holes of the ensemble approximate the intersection of porosities when mistakes differ, and approximate the first porosity when they align. Prefer adding a different-porosity model, including a low-quality one, over adding a high-quality copy of the same family.

DEFINITIONS

[0006] As used herein, "Family" means An occupancy class: training house plus tool surface plus observed failure mode. Not a clustering theorem.

[0007] As used herein, "Porosity" means A hole-set measure of a coder (error classes / residual miss surface). Exact metric is an embodiment.

[0008] As used herein, "Different-mistake overlay" means Seating models whose hole-sets are not aligned, including a low-quality model of another family.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a schematic of two hole-sets (porosities). Same-family copies leave holes aligned. A different-mistake overlay covers residual surface faster than stacking copies. Reason 1990 swiss-cheese is an analogy, not a claimed accident-causation theory.

[0010] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0011] Assign each model a family. Occupancy rule, not a clustering theorem.

[0012] Quantify each coder's porosity $p(m)$ as a hole-set measure. Embodiments include pass/fail per task class and co-failure beta. Do not put a fake number in this specification.

[0013] When seating votes (architecture, build, critics): the same family does not count twice.

[0014] Overlay: residual holes of the ensemble approximate the intersection of porosities when mistakes differ; they approximate the first porosity when they align.

[0015] Prefer adding a different-porosity (different-mistake) model, including a low-quality one, over adding a high-quality copy of the same family. Predicted: surface cover grows faster than the same-family baseline (flat, or linear only in invoice).

[0016] Dual-lane (COSMOS-P02) and different-family critics (COSMOS-P05) exist to punch different holes.

[0017] Hypothesis to measure, not a number in this provisional: coverage versus N for different-family overlays versus same-family copies. Same-family incremental coverage approximates zero. Different-mistake overlay: residual approximates the product of unaligned hole probabilities; cover of the remaining surface is large on the first different overlay.

[0018] Swiss-cheese is an image (Reason 1990 as analogy). This disclosure does not claim Reason's accident-causation theory as the invention.

BEST MODE

[0019] Encoded docs/MOTIF.md 2026-09-07: family-axis occupancy; dual-lane plus different-family critics; named pins rather than a rotating free-model router (silent swap is a hole that looks like coverage).

FURTHER EMBODIMENTS

[0020] A coding critic seat uses a Google-family model and an OpenRouter value coder of another family, not two copies of one API.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0021] Swiss-cheese occupancy language is in MOTIF.md on public cosmos (method text 2026-08-25; swiss cheese 2026-09-07 still in-tree). Family-axis papers of 2026 are their measurements, not this inventor's.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0022] Must name: Reason 1990 / BMJ 2000 (metaphor); Knight and Leveson 1986; Hidden Clones arXiv:2603.17111; LLMs-as-Jury arXiv:2607.10139; co-failure ceiling arXiv:2606.27288; Wisdom and Delusion of LLM Ensembles arXiv:2510.21513; More Agents Is All You Need arXiv:2402.05120 (anti-shape: same-model copies); Self-consistency arXiv:2203.11171; Mixture-of-Agents arXiv:2406.04692; Self-MoA arXiv:2502.00674 (mixing can hurt); Estornell and Liu NeurIPS 2024; DZone/Cognaptus swiss-cheese AI as safety-stack metaphor.

[0023] Patents on this point are thin. No United States patent found that claims same family is one vote or porosity overlay. Unpublished applications are unknown.

WHAT THIS DISCLOSURE IS NOT

[0024] Not Reason's aviation model. Not ensembles exist. Not a measured superlinear exponent. Not a clustering theorem.

STATEMENT OF INVENTION (NOT CLAIMS)

[0025] Quantified porosity of each coder; seating and overlay rules so that different-mistake (including low-quality) models cover residual surface faster than same-family copies.

[0026] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0027] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.